



Resin-Based Stereolithography for Pre-Production Footwear Prototyping: A Narrative Review and Workflow Framework for Indian Footwear Development

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Abstract: Footwear development requires repeated verification of geometry, dimensional conformity, component interfaces, surface details, and manufacturing constraints before production tooling is finalized. Resin-based stereolithography (SLA) offers high geometric resolution and surface quality, but its role in pre-production footwear development remains less clearly defined than its use in general additive manufacturing (AM). This study examines the applications, technical limitations, and industrial relevance of resin-based SLA as a physical prototyping method for footwear design and manufacturability validation, with particular consideration of the Indian footwear sector. A structured narrative review was conducted using English-language literature published between January 2016 and August 2025. Relevant publications were identified through Scopus, Web of Science, PubMed, and Google Scholar. Studies directly addressing footwear were prioritized, while evidence from related AM and engineering applications was included when it informed footwear prototype validation. The reviewed evidence showed that SLA reproduced complex sole contours, arch geometries, lattice structures, textured surfaces, and assembly features with fine geometric detail. Compared with fused deposition modelling (FDM), SLA generally provided finer feature resolution and smoother surfaces under the reported processing conditions. Its accuracy and mechanical response depended strongly on resin formulation, layer thickness, build orientation, printer configuration, support strategy, and post-curing conditions. SLA prototypes also supported the preliminary inspection of clearances, undercuts, wall thicknesses, bonding regions, and component interfaces before tooling. However, photopolymer resins did not reliably reproduce the flexibility, viscoelasticity, fatigue resistance, or long-term behaviour of production footwear materials. The evidence was further limited by small samples, inconsistent process reporting, limited anthropometric diversity, and a lack of longitudinal validation. Based on these findings, a workflow framework was developed to connect digital design, SLA fabrication, physical inspection, manufacturability assessment, and iterative design revision. Resin-based SLA is best positioned as a high-fidelity design-verification and pre-tooling assessment method rather than a substitute for functional testing of finished footwear. The proposed framework provides a practical basis for integrating SLA into footwear development in India while identifying the validation required before broader industrial use.

Keywords: Resin-based stereolithography; Footwear prototyping; Digital fabrication; Design verification; Dimensional accuracy; Manufacturability assessment; Pre-production validation; Indian footwear industry

1 Introduction

1.1 Background

The global footwear industry represents a major economic sector with a market value of hundreds of billions of dollars. In an increasingly competitive market, footwear manufacturers face continuing pressure to improve product design, functional performance, manufacturing efficiency, and environmental performance. Footwear development conventionally proceeds through several iterative stages, including handcrafted model preparation, computer numerical control machining of molds, prototype fabrication, and physical testing. These stages can

require considerable time, labour, and material resources, particularly when design changes are identified late in the development process and further prototype or tooling iterations become necessary [1, 2].

The adoption of additive manufacturing (AM) has altered conventional footwear prototyping by allowing physical components to be fabricated directly from computer-aided design (CAD) models through the successive deposition or solidification of material. This digital-to-physical route can shorten the interval between design and prototype evaluation, reduce waste associated with material removal, and permit the fabrication of complex geometries that are difficult or impractical to produce by conventional methods [3, 4]. These characteristics are relevant to footwear components containing curved anatomical surfaces, variable wall thicknesses, textured regions, internal features, or lattice-based structures.

Among vat photopolymerization processes, stereolithography (SLA) has attracted particular attention because of its dimensional accuracy, fine-feature resolution, and relatively smooth surface finish. SLA uses ultraviolet (UV) light, commonly delivered by a controlled laser or comparable light source, to selectively cure liquid photopolymer resin and construct a three-dimensional part layer by layer. Under suitable printing and post-processing conditions, reported feature dimensions are approximately 25–100 μm , while surface roughness values below 2 μm have been achieved in some applications [5, 6]. These values are not universal process specifications, because the outcome depends on the printer, resin system, component geometry, build orientation, layer thickness, and post-curing procedure.

The resolution and dimensional control associated with SLA have led to its use in prototype development across aerospace, automotive, consumer-product, and biomedical applications [7]. In footwear development, the process has been investigated for producing detailed physical representations of midsoles, uppers, outsoles, and related sole components. Such prototypes allow complex features—including sole contours, lattice architectures, variable-thickness profiles, surface textures, and ergonomic geometries—to be examined before a design advances to tooling or another resource-intensive manufacturing stage [1, 5, 6, 8].

Developments in tough, elastic, and flexible photopolymer formulations have extended SLA beyond the production of rigid visual models. These resins can be used to prepare semi-functional prototypes for preliminary examination of flexibility, deformation, fit, or selected load-transfer concepts. Their use nevertheless requires a clear distinction between prototype-level assessment and validation of the final product. The mechanical, viscoelastic, and fatigue properties of an SLA resin cannot be assumed to represent those of the elastomeric and polymeric materials used in production footwear without material-specific testing [9].

Since its commercialization by Charles (“Chuck”) Hull in the 1980s, SLA has undergone substantial development in printer architecture, build volume, curing control, dimensional accuracy, and photopolymer formulation. These advances have contributed to the wider technical maturation of AM and expanded the range of geometries and applications that can be addressed through vat photopolymerization [10]. They are particularly pertinent to footwear development, where shorter development cycles, repeated design iteration, customization, dimensional consistency, and efficient use of resources are important production considerations.

AM-based prototyping can also reduce the material removal associated with conventional subtractive tooling and model-making operations. This reduction should not, however, be treated as sufficient evidence of lower environmental impact. The sustainability of SLA depends on resin chemistry, printing energy, material utilization, washing and post-curing requirements, waste handling, and end-of-life treatment [11]. An assessment of SLA in footwear development must therefore consider both its potential contribution to more efficient design iteration and the environmental limitations of photopolymer-based fabrication.

Despite its advantages in precision and surface reproduction, SLA has received less attention in footwear research than selective laser sintering (SLS) and fused deposition modelling (FDM) in several application areas. This imbalance is especially apparent in studies of end-use footwear production and compliant lattice midsoles, where powder-bed fusion and material-extrusion processes are often selected because of their available production materials and mechanical characteristics [1, 2, 4, 12]. The lower prevalence of SLA in end-use production does not diminish its relevance at the pre-production stage. Its principal value lies in generating detailed physical representations of digital footwear designs for geometric inspection, dimensional verification, surface assessment, assembly checks, and iterative design revision. This role provides the basis for examining SLA as a link between digital footwear design and production planning, including its possible integration into the varied manufacturing conditions of the Indian footwear sector.

1.2 Design Engineering Relevance

Footwear components commonly contain intricate surfaces, localized structural features, variable wall thicknesses, lattice geometries, and closely interacting interfaces. These characteristics make physical verification important even when a design has already been assessed in a digital environment. Early inspection of geometry and aesthetic details can reveal dimensional, structural, or assembly-related problems before molds and other dedicated tooling are produced. Conventional prototype preparation may be slow or costly when the component requires complex

geometry, several design iterations, or assessment of the interaction between form and material behaviour.

Resin-based SLA creates high-resolution physical representations directly from digital models and can therefore occupy an intermediate position between CAD-based analysis and production tooling. The fabricated model can be examined visually, measured dimensionally, handled physically, and assembled with adjoining components. These activities support geometric, surface, dimensional, and assembly-oriented assessment before the design progresses to subsequent manufacturing stages [12–15]. The physical prototype does not replace digital analysis; rather, it allows assumptions made in the digital model to be checked against a fabricated object.

SLA prototypes can also support quantitative or semi-quantitative examination of heel-to-toe geometry, component dimensions, midsole profiles, localized curvature, and conformity among the upper, last, and sole assembly. When used together with a footwear last, the prototype can help designers inspect dimensional compatibility and identify fit-related geometric discrepancies during an early design iteration [16]. This form of assessment concerns conformity between designed components and intended dimensions. It should be distinguished from wearer comfort or biomechanical performance, which depends on production materials, complete footwear construction, loading conditions, foot morphology, and gait.

From a manufacturing perspective, a physical SLA model can be used before tooling to inspect draft requirements, undercuts, parting-line locations, wall thicknesses, component interfaces, and assembly clearances. It can also support an initial assessment of snap-fit concepts, bonding regions, component alignment, and other interface features before resources are committed to production molds or tooling [17, 18]. Such inspection can expose design constraints that are difficult to judge from screen-based visualization alone and can complement digital design for manufacturability (DFM) analysis. Identifying these problems during physical prototyping reduces the likelihood that dimensional or assembly errors will first become apparent after the production tooling has been completed. It can therefore limit the time, cost, and material consequences of late design changes. This capability is relevant to performance-oriented and premium footwear, where precise geometry, rapid iteration, customization, surface definition, and product differentiation can influence commercial competitiveness [19]. At this stage, the main engineering role of resin-based SLA is thus to convert digital footwear geometry into a high-resolution physical model that supports early verification and better-informed manufacturing decisions.

1.3 Summary of Existing Evidence

Research on AM in footwear has expanded in recent years and has addressed personalization, sustainable production, rapid prototyping, product customization, and lattice-based footwear structures [1, 3, 20]. Within this wider field, resin-based SLA has shown particular value in producing detailed prototypes of complex sole geometries, surface features, aesthetic elements, and other components for which geometric definition and surface fidelity are important [5, 6].

Compared with extrusion-based FDM, SLA generally produces finer features and smoother surfaces under the processing conditions reported in the literature. These characteristics make it suitable for examining textures, branding elements, contour transitions, and dimensional details during design verification [1, 3, 6]. The comparison is application-dependent: SLA is not necessarily the preferred process when prototype cost, build size, material flexibility, impact resistance, or long-term mechanical loading is the primary concern.

Flexible and tough photopolymer resins have also been investigated in prototypes intended for preliminary assessment of component flexibility, fit-related geometry, and load-transfer concepts [6, 10, 17, 18]. Such prototypes can contribute useful information during design development, but they remain engineering assessment models unless their material response has been validated against the intended production material. Apparent geometric fidelity or short-term deformation behaviour does not establish equivalence in cushioning, fatigue resistance, energy return, wearer comfort, or durability.

Relative to SLS and FDM, SLA is particularly suited to applications that place a high priority on intricate geometry, dimensional detail, and surface reproduction. Process selection must nevertheless be based on the purpose of the prototype, the required material behaviour, the relevant manufacturing constraints, and the type of evidence needed for the next design decision [1, 6, 18]. Taken together, the available literature indicates that SLA can assist rapid design iteration, dimensional inspection, surface assessment, and the identification of pre-production design problems. Evidence from the wider AM field also indicates that digitally fabricated prototypes can shorten development cycles and improve the efficiency of repeated prototype evaluation [4, 21].

1.4 Limitations and Inconsistencies in Current Literature

The literature on SLA-based footwear prototyping remains methodologically inconsistent. Reported tensile strengths, which extend approximately from 30 to 100 MPa, cannot be compared directly when resin chemistry, specimen geometry, build orientation, layer thickness, exposure settings, and post-curing conditions differ among studies. Inadequate control or reporting of these factors also obscures their contribution to anisotropy and mechanical

response [5, 7]. Numerical ranges reported for individual resins or test configurations should therefore not be interpreted as general performance limits for SLA footwear prototypes.

Biomechanical investigations have often relied on short-duration tests, simplified loading conditions, or proxy specimens fabricated from photopolymers whose behaviour differs from that of common production materials such as ethylene-vinyl acetate (EVA) and thermoplastic polyurethane (TPU). Consequently, the extent to which results obtained from an SLA prototype predict the response of a finished footwear component remains uncertain [8, 22]. This distinction is particularly important where a prototype is used to discuss cushioning, plantar-pressure redistribution, flexibility, fatigue resistance, or long-term wearer performance.

Manufacturing-oriented evidence is also limited. Draft-angle inspection, undercut detection, assembly trials, interface verification, and other pre-tooling applications have received less attention than the direct manufacture of end-use AM components [1, 13]. As a result, the literature does not yet establish a consistent method for determining which physical checks should be conducted before a footwear design moves from CAD modelling to tooling.

Small experimental samples, heterogeneous test procedures, and incomplete reporting of resin formulation and process variability further restrict reproducibility and meaningful comparison across studies [6]. Sustainability evidence is similarly incomplete, particularly regarding the environmental burden of photopolymer resins, consumables used during cleaning and post-processing, resin disposal, and the availability of bio-based formulations suitable for footwear prototyping [12]. The existing evidence is therefore useful but dispersed across material, process, design, and application studies. It does not yet supply a coordinated framework for integrating digital design, SLA fabrication, physical inspection, manufacturability assessment, and design revision within a pre-production footwear workflow [3].

1.5 Knowledge Gap and Objectives

Existing reviews of AM in footwear have concentrated mainly on production technologies, inlaid soles and the direct customization of footwear geometry and size. The more specific role of SLA as a pre-production verification method has received substantially less attention [1, 3]. In particular, the literature has not clearly established how the dimensional precision and surface definition of SLA can be used to verify footwear geometry, examine manufacturability, and identify tooling-related risks before production decisions are finalized [1].

Design validation, dimensional accuracy, fit-related geometry, manufacturing constraints, and tooling decisions have generally been examined as separate topics rather than as connected stages of a single pre-production process [2, 4]. Consequently, designers and manufacturers have limited guidance on how evidence from an SLA prototype should inform the transition from a CAD concept to a production-ready footwear design. This gap is consequential because errors identified after mold preparation can lead to repeated tooling work, longer development schedules, additional material use, and higher manufacturing costs.

This narrative review therefore examines resin-based SLA as a physical prototyping and pre-production validation approach for footwear development. It synthesizes evidence concerning geometric and dimensional verification, fit-related geometry, surface and aesthetic assessment, material limitations, manufacturability evaluation, component interfaces, and tooling-risk reduction. Evidence from direct footwear studies is considered together with transferable findings from related AM and engineering applications. Particular attention is given to the relevance of this evidence to the Indian footwear sector, while recognizing that the literature search is not geographically restricted and that industrial applicability must be established under local production conditions. Based on the synthesis, the review develops a workflow framework linking user and market data, CAD modelling, digital analysis, SLA fabrication and post-processing, physical inspection, manufacturability assessment, iterative design revision, and pre-production planning. The framework is intended to clarify where SLA can support design and tooling decisions, where complementary digital or experimental methods remain necessary, and why SLA prototypes should not be treated as substitutes for validation of the mechanical or biomechanical performance of finished footwear.

2 Methodology

This article was conducted as a narrative review with a structured literature search to synthesize and critically discuss the translational applications of resin-based SLA in pre-production physical prototyping for footwear development. A narrative approach was considered appropriate because the available literature spans multiple disciplines, including AM, materials engineering, footwear design, and manufacturing. Publications were selected based on their relevance to the objectives of this narrative review. Eligible literature addressed resin-based SLA or related technologies and their applications in functional prototyping, design validation, dimensional accuracy, material performance, manufacturability, or footwear development. Studies with direct footwear applications were prioritized, while evidence from other engineering and AM applications was considered when it provided transferable knowledge relevant to pre-production footwear validation. Duplicate records were identified by comparing publication titles, authors, publication years, sources, and DOIs, where available, and were removed before screening. The remaining records were assessed by title and abstract, followed by full-text evaluation of

potentially relevant publications. As this article was conducted as a narrative review rather than a systematic review, formal systematic-review procedures and PRISMA-based study selection were not applied.

The literature search included English publications from January 2016 to August 2025 (Table 1). Relevant studies were identified through Scopus, Web of Science, PubMed, and Google Scholar using keyword combinations and Boolean operators. Database-specific search strategies were developed for each platform (Table 2).

Table 1. Search limits across databases

Parameter	Applied Criteria
Publication period	January 2016–August 2025
Language	English
Document focus	Peer-reviewed journal articles, relevant conference proceedings, and technically relevant reports
Geographical restriction	None
Primary research domains	Additive manufacturing (AM), materials engineering, footwear engineering, biomechanics, ergonomics, product design, and manufacturing
Additional screening terms	Functional prototyping, tooling risk reduction, biomechanical simulation, lattice structures, mechanical performance, and manufacturability

Table 2. Literature search strategy across selected databases

Database	Primary Topic/Search Focus	Search Terms/Strategy
Scopus	SLA technology and footwear applications	TITLE-ABS-KEY (“SLA” OR “resin-based AM” OR “vat photopolymerization”) AND TITLE-ABS-KEY (“footwear” OR “shoe” OR “sole” OR “midsole” OR “insole” OR “last”)
Scopus	Functional prototyping and design validation	TITLE-ABS-KEY (“functional prototyping” OR “prototype validation” OR “product validation” OR “design validation” OR “pre-production validation”) AND TITLE-ABS-KEY (“SLA”)
Scopus	Manufacturability and tooling validation	TITLE-ABS-KEY (“manufacturability assessment” OR “design for manufacturing” OR “tooling risk reduction” OR “draft angle” OR “undercut” OR “mold flow”) AND TITLE-ABS-KEY (“AM” OR “SLA”)
Web of Science	SLA-based footwear design and customization	TS = (“SLA” OR “resin-based AM”) AND TS = (“footwear” OR “shoe” OR “sole” OR “insole” OR “customized last”)
Web of Science	Lattice structures and functional performance	TS = (“lattice structure” OR “lattice generation” OR “cellular structure”) AND TS = (“footwear” OR “sole” OR “midsole” OR “AM”)
PubMed	User fit and ergonomic performance	(“footwear” OR “insole” OR “orthotic device”) AND (“comfort” OR “fit” OR “ergonomics” OR “plantar pressure” OR “gait”)
PubMed	Translational applications of AM	(“3D printing” OR “AM” OR “SLA”) AND (“orthotic” OR “insole” OR “footwear” OR “biomechanical”)
Google Scholar	Broad footwear prototyping literature	“SLA footwear prototyping”; “SLA footwear design”; “resin 3D printing footwear prototypes”
Google Scholar	Pre-production validation and manufacturing	“Functional prototyping footwear”; “pre-production footwear validation”; “design validation AM footwear”; “manufacturability assessment footwear prototypes”
Google Scholar	Mechanical and material performance	“SLA resin mechanical properties footwear”; “energy absorption 3D printed lattice footwear”; “cyclic fatigue resin AM”; “wear resistance 3D printed polymers”
Google Scholar	Emerging and interdisciplinary applications	“Biomechanical simulation footwear AM”; “lattice midsole AM”; “digital footwear customization”; “SLA functional prototype validation”

Note: AM = additive manufacturing; SLA = stereolithography; TS = topic search; TITLE-ABS-KEY means title, abstract, and keywords.

3 Thematic Literature Synthesis

3.1 Fundamentals of Resin-Based Stereolithography in Footwear Prototyping

Resin-based SLA is capable of producing footwear prototypes with high geometric resolution and fine surface detail, making it particularly suitable for evaluating complex component geometries, lattice structures, and aesthetic features [5, 6].

In footwear applications, SLA-generated lattice structures may be useful for the preliminary assessment of geometric concepts intended to influence cushioning or energy management. However, the mechanical response of photopolymer resins may differ substantially from that of production materials, and therefore SLA prototypes should not be assumed to reproduce the long-term elasticity, fatigue behaviour, or viscoelastic response of materials such as EVA or TPU [1, 9].

Printing parameters, particularly layer thickness and build orientation, can influence dimensional accuracy and mechanical properties. Finer layer thicknesses, such as 25–50 μm , can improve the reproduction of small geometric features and surface details but generally increase build time. Build orientation may also contribute to anisotropic mechanical behaviour, particularly in thin-walled or lattice-based structures, and reported differences in compressive properties across orientations highlight the importance of controlling and documenting printing conditions [2, 4].

Accordingly, comparisons of mechanical performance between studies should account for layer thickness, build orientation, resin formulation, printing parameters, and post-processing conditions. Experimental approaches commonly involve fabrication of standardized specimens using desktop or laboratory-scale SLA systems, followed by mechanical characterization according to relevant standards, such as ASTM D638 for tensile testing [2, 3, 6].

Sample sizes are often relatively small, and post-curing procedures may vary in duration and exposure conditions. Such variations can substantially influence the final properties of photopolymerized parts and limit direct comparison between studies. Computational approaches, including finite element modelling, can additionally be used to investigate stress and deformation distributions within prototype geometries; however, model predictions depend on the accuracy of the material properties and boundary conditions used and may not fully capture printing-induced defects, polymerization effects, or dimensional changes during processing [2, 3, 6].

The high surface quality achievable with SLA can support visual, dimensional, and aesthetic evaluation, including assessment of surface texture and fine design details [4, 7]. Nevertheless, reported dimensional tolerances vary according to the printer, resin, geometry, build orientation, and processing conditions, making generalized tolerance values difficult to establish across different systems [2, 5].

Small sample sizes and inconsistent reporting of resin formulation and post-processing further limit reproducibility and cross-study comparisons. Two important gaps remain in the current evidence base. First, SLA prototyping studies do not consistently incorporate footwear-specific anthropometric requirements, including variation in foot dimensions, shape, and fit. Second, findings obtained from individual laboratory-scale geometries may have limited generalizability across the wide range of footwear sole designs and intended functional applications [1, 3].

3.2 Design Validation in Footwear Engineering

Previous studies indicate that resin-based SLA can provide high geometric fidelity for footwear components, particularly when reproducing complex contours around the medial and lateral arches, heel regions, and other anatomically relevant surfaces [5, 8, 9].

The dimensional accuracy achievable with SLA may facilitate early verification of design parameters such as heel-to-toe geometry, component thickness, curvature, and interface alignment before progression to production tooling. However, reported accuracy is dependent on factors including printer resolution, build orientation, resin formulation, geometric complexity, and post-processing conditions; therefore, accuracy values should be interpreted within the specific experimental context rather than generalized across all SLA systems. Some studies have also reported associations between SLA-based prototype geometries and simulated or experimentally assessed pressure-distribution patterns, with correlation coefficients reported in the range of approximately 0.80–0.92 [5, 8, 9].

Such findings suggest that physical prototypes may be useful for preliminary evaluation of design-dependent load-distribution concepts. Nevertheless, correlation with a pressure model should not be interpreted as evidence that an SLA prototype reproduces the biomechanical behaviour of a finished footwear product. Pressure distribution is influenced by multiple factors, including material compliance, foot morphology, loading conditions, footwear construction, and user-specific gait characteristics.

Consequently, biomechanical performance should be verified using appropriately instrumented experimental testing rather than inferred from geometric agreement alone. Fit-related assessments further suggest that SLA can provide useful physical representations of digitally generated footwear geometries and foot-scan-derived designs [2, 11].

This enables designers to identify potential discrepancies between the intended digital geometry and the physical prototype before investing in production tooling. SLA may therefore be particularly valuable for evaluating

last-upper-sole interfaces, dimensional conformity, component positioning, and localized geometric modifications during iterative development. Consequently, a geometrically accurate SLA prototype may not provide a reliable indication of actual wearer comfort or long-term functional performance [2, 11].

This distinction is important because “geometric validation and ergonomic validation represent complementary but different stages of footwear development”. Available evidence supports SLA primarily as an “early-stage design-verification and physical validation tool”. Its principal engineering value lies in enabling rapid assessment of geometry, dimensional conformity, interfaces, and manufacturability before expensive tooling or production trials.

Where the intended design depends on cushioning, flexibility, pressure redistribution, or other biomechanical functions, SLA prototyping should be followed by material-specific mechanical characterization and appropriate design. This staged approach can strengthen the translational pathway from digital footwear design to validated production-ready components while reducing the risk of premature decisions based solely on geometric prototype fidelity.

3.3 Manufacturability Assessment Before Tooling

Manufacturability assessment is an important stage in footwear development because design-related errors identified after production tooling can result in substantial additional time, cost, and material consumption. In this context, resin-based SLA can provide a practical physical verification step between digital design and production tooling. High-resolution SLA prototypes can be used to inspect geometrical features that may influence subsequent manufacturing, including undercuts, draft requirements, parting-line locations, wall thickness, clearances, and component interfaces [4, 6, 14].

Such assessments can complement computer-aided manufacturability analysis by allowing designers to examine the physical representation of the proposed geometry before committing to molds or other dedicated tooling. Physical prototypes can additionally support preliminary evaluation of assembly-related features, including bonding interfaces, component alignment, snap-fit concepts, and dimensional clearances [4, 6]. Dimensional measurements using tools such as digital calipers can be combined with visual and tactile inspection to identify discrepancies between the digital design and fabricated prototype. However, the dimensional accuracy of SLA components is influenced by resin formulation, printer configuration, build orientation, layer thickness, shrinkage, and post-processing; therefore, prototype measurements should be interpreted in relation to the specific manufacturing conditions [14, 16].

The literature also indicates that virtual simulation and physical prototyping provide complementary forms of manufacturability assessment. Software-based approaches can evaluate geometric constraints and, depending on the modelling framework, investigate parameters such as material flow, deformation, or assembly behaviour before fabrication. Physical SLA prototypes, in contrast, provide direct evidence of the fabricated geometry and can reveal practical issues associated with surface condition, tolerances, assembly interfaces, and post-processing that may not be fully represented in digital models [14, 16].

Neither approach should therefore be considered sufficient in isolation for comprehensive manufacturing validation. Mechanical testing can further strengthen pre-tooling assessment when the intended component is expected to withstand functional loading. Compression or other component-specific tests can be used to evaluate structural response under defined loading conditions; however, the selected load range and test configuration should reflect the intended application and be reported consistently across studies. Results obtained from SLA prototypes should also be interpreted cautiously because photopolymer resins may differ substantially from production materials in stiffness, toughness, fatigue resistance, and deformation behaviour [14, 16].

A further limitation of the existing evidence is the predominance of relatively simple geometries, individual components, or single-resin systems, with comparatively limited investigation of multipart footwear assemblies and complex component interactions [13, 15]. This may restrict the generalizability of reported manufacturability findings. Similarly, small numbers of prototype iterations can provide useful qualitative information but are insufficient to establish reliable defect rates or generalized failure probabilities [4, 6].

Overall, the available evidence supports SLA as a “pre-tooling design-verification platform rather than a substitute for production-tool validation”. Its principal contribution is the early identification of geometric, dimensional, and assembly-related issues through rapid physical iteration. A robust workflow would combine digital manufacturability analysis with SLA-based physical verification and, where relevant, material-specific mechanical testing before final tooling. This staged approach can reduce the likelihood of late-stage design modifications while providing a more evidence-based transition from digital footwear concepts to production-ready designs [1, 18].

Figure 1 illustrates the integrated workflow for resin-based SLA footwear prototyping, physical validation, and pre-production development. The process progresses from market and user-data collection through data interpretation, parametric CAD modelling and digital simulation, SLA 3D printing and post-processing, high-fidelity physical prototype generation, physical and manufacturability validation, and finally design-for-manufacturing and pre-production planning. The validation stage incorporates dimensional verification, functional evaluation, draft-angle and undercut detection, wall-thickness and rib optimization, and snap-fit and assembly compatibility

assessment. Solid arrows indicate sequential process flow, while dashed arrows indicate iterative feedback and design refinement. This framework demonstrates how resin-based SLA prototyping can support the transition from digital footwear design to validated pre-production and subsequent mass manufacturing.

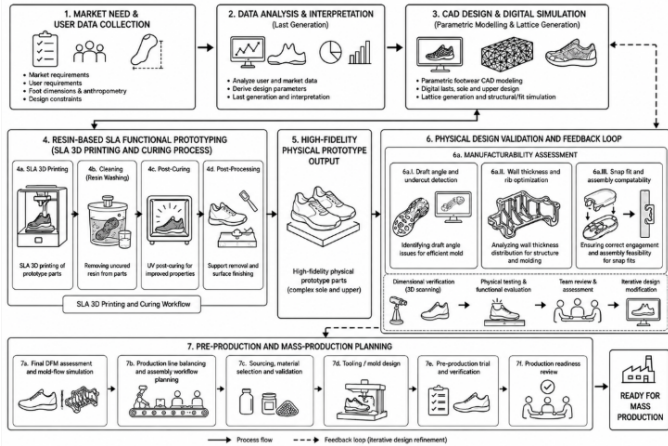


Figure 1. Integrated workflow for resin-based SLA footwear prototyping and pre-production development

Note: SLA = stereolithography; CAD = computer-aided design; UV = ultraviolet; DFM = design for manufacturability.

3.4 Comparative Analysis: Resin-Based Stereolithography vs. Other Additive Manufacturing Technologies in Footwear Prototyping

Comparative evidence indicates that the principal advantage of resin-based SLA in footwear prototyping is its high geometric resolution and surface quality. Under appropriate processing conditions, SLA can reproduce fine contours, textures, and intricate lattice features more accurately than many FDM systems, although dimensional accuracy and surface roughness depend on the specific printer, resin, orientation, and processing parameters [1, 2].

FDM generally offers lower material and equipment costs and can be advantageous for rapid, economical prototypes, but layer-wise deposition may result in more visible surface features and lower resolution for fine geometries [3]. SLS, in contrast, can provide greater suitability for mechanically demanding prototypes because of its broader range of engineering-grade powders and absence of conventional support structures. This makes SLS particularly relevant to functional and load-bearing footwear components, although its surface finish and fine-detail reproduction may differ from those achievable with SLA.

SLA can therefore be advantageous when the primary objective is geometric, aesthetic, dimensional, or interface validation, particularly in applications where accurate shape reproduction, fine surface details, and dimensional consistency are more critical than long-term mechanical durability or fatigue resistance. However, the overall economic feasibility of SLA remains highly dependent on specific manufacturing conditions. Reported cost comparisons vary substantially due to differences in material consumption, machine type, post-processing requirements, build volume utilization, and production scale, which makes it difficult to establish universally applicable cost estimates across different application scenarios [3].

Importantly, comparative studies often emphasize different performance criteria depending on the characteristics and intended applications of each AM technology. SLA investigations frequently prioritize factors such as printing resolution, surface quality, and dimensional accuracy, whereas FDM and SLS studies may place greater emphasis on material durability, mechanical strength, and long-term performance characteristics [2]. This methodological variation in evaluation priorities introduces potential selection bias in cross-technology comparisons, as differences in assessment criteria may influence the interpretation of relative advantages and limitations among these technologies.

Overall, current evidence suggests that “SLA is particularly valuable for high-fidelity pre-production design validation, FDM for economical and rapid iteration, and SLS for more mechanically oriented prototyping”. The most appropriate technology should therefore be selected according to the intended validation objective rather than on the basis of a single performance metric [1].

Table 3 summarizes the representative findings reported under the specific conditions of the cited studies and should not be interpreted as standardized performance specifications or universally applicable benchmarks. Variations in material formulation, printer configuration, build orientation, layer thickness, component geometry, post-processing procedures, and testing methodologies may significantly influence the reported outcomes, including dimensional accuracy, surface quality, mechanical behaviour, processing requirements, and overall cost. Therefore, cross-technology comparisons should be interpreted cautiously, as differences in experimental conditions and evaluation criteria may affect the direct comparability of the reported results.

Table 3. Comparative overview of different criteria for SLA, FDM, and SLS in the context of pre-production footwear prototyping

Criterion	SLA (Resin-Based)	FDM (Filament)	SLS (Powder-Based)	Footwear Prototyping Implication	References
Surface finish (Ra)	Reported values < 2 μm in some studies	Visible layer patterns; reported values vary with layer height and process parameters	Typically textured/grainy surface; values vary with powder and processing conditions	SLA may be advantageous for aesthetic, texture, and fine-feature assessment	[1, 5, 6]
Dimensional accuracy	Approximately ± 0.05 – 0.15 mm in selected studies	Approximately ± 0.1 – 0.3 mm under specific conditions	Approximately ± 0.1 mm in selected studies	SLA may be useful for detailed geometry, contour, and fit-related checks; direct comparisons depend on test conditions	[1, 6, 17, 18]
Mechanical fidelity	Generally suitable for short-term prototype assessment, depending on resin; some formulations may be relatively brittle	Can provide functional prototypes; performance depends on filament and print parameters	Often suitable for mechanically functional prototypes; performance depends on powder and processing	SLS may be more appropriate when mechanical loading is a primary validation objective, whereas SLA is useful for geometric assessment	[5, 6, 17, 18]
Post-processing	Typically requires support removal and washing/UV post-curing	Usually limited to support removal and surface finishing, depending on geometry	Requires removal/recovery of residual powder and possible finishing	Post-processing requirements vary by geometry and system; FDM may involve fewer finishing steps in some applications	[2, 4, 6]
Suitability for midsole prototyping	High geometric detail; limited evidence for long-term fatigue behaviour	Applicable for selected prototype geometries; mechanical performance depends on material	Potentially suitable for mechanically functional or load-bearing prototypes	SLA may be useful for early geometric/lattice assessment, whereas technologies with more suitable mechanical properties may be preferable for load-related evaluation	[1, 6, 17, 18]
Suitability for upper prototyping	High detail and potential for interface/fit assessment	Applicable depending on geometry and material	Applicable for selected geometries	SLA may be particularly useful when detailed geometry, surface features, or interface conformity are the primary objectives	[1, 5, 18]
Cost per prototype	Approximately USD 10–50 reported in selected studies	Approximately USD 5–20 reported in selected studies	Approximately USD 20–100 reported in selected studies	Prototype cost is highly dependent on material, machine, geometry, build volume, post-processing, and production scale; FDM may be less expensive in some settings	[1, 3, 6]

Note: SLA = stereolithography; FDM = fused deposition modelling; SLS = selective laser sintering; UV = ultraviolet; USD = United States Dollar.

In Figure 2, representative prototype models shown in panels A, B, and C were used to evaluate dimensional accuracy, geometric conformity, and physical design characteristics during the preliminary validation stage before subsequent design refinement and pre-production development.

These evaluations focused on examining whether the fabricated prototypes accurately reflected the original digital models in terms of overall shape, structural features, and key dimensional parameters. By comparing the physical prototypes with the intended digital designs, the validation process provided a basis for identifying potential discrepancies and supporting iterative improvements in the design process. Such prototype-based assessments contributed to enhancing the consistency between digital concepts and manufactured models. Through the integration of digital design verification and physical prototype evaluation, the preliminary validation stage served as a transitional step between conceptual modelling and subsequent production-oriented development.

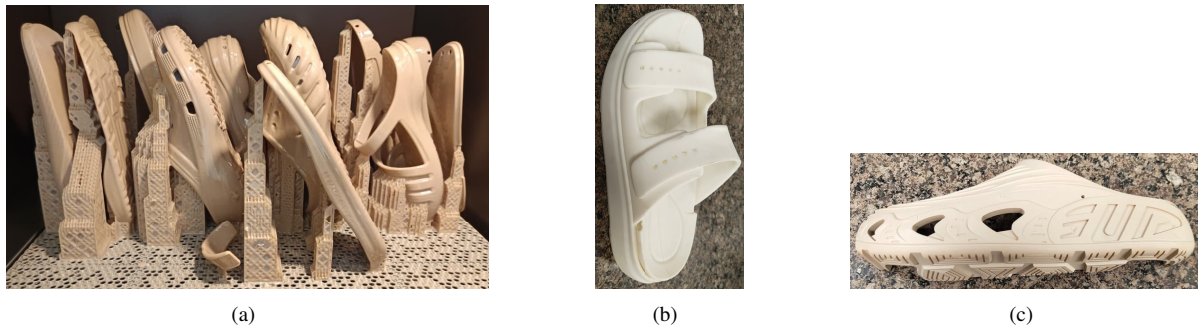


Figure 2. Resin-based 3D-printed footwear prototypes for physical design validation: (a) prototype A; (b) prototype B; (c) prototype C

4 Critical Analysis & Knowledge Gaps

4.1 Methodological Weaknesses in Current Literature

The current literature on resin-based SLA for footwear prototyping remains methodologically fragmented. Many studies examine individual variables, such as layer thickness, build orientation, or resin formulation, without systematically integrating these factors into complete pre-production workflows [2, 5, 8]. Although variable-isolation studies are useful for understanding individual effects, they may not adequately capture interactions among processing parameters, including the combined influence of curing conditions, polymerization shrinkage, build orientation, and geometric complexity on dimensional accuracy and structural response. A further limitation is the relatively limited integration of experimental testing with computational modelling. Hybrid experimental-simulation approaches could provide a more comprehensive understanding of prototype behaviour and potentially improve the reliability of design decisions during iterative development [1, 3]. Evidence derived from laboratory-scale desktop SLA systems should also be interpreted cautiously, as parameters optimized for small prototype sections may not directly translate to larger or more complex footwear assemblies [4, 6]. Differences in build volume, thermal and curing conditions, support configuration, and geometric scale may influence dimensional and mechanical outcomes. Moreover, many studies remain focused on proof-of-concept demonstrations rather than iterative engineering workflows incorporating repeated design–prototype–test–redesign cycles. Limited reporting of such feedback loops reduces the translational relevance of existing evidence to commercial footwear development. Consequently, future research should prioritize standardized experimental protocols, multi-factorial designs, validated simulation methods, and application-scale iterative testing to establish more robust evidence for SLA-based pre-production footwear prototyping.

4.2 Small Sample Bias

Small sample sizes remain an important limitation in studies evaluating SLA-based footwear prototypes. Prototype studies frequently involve only a small number of fabricated specimens, while participant-based fit or ergonomic assessments may also include relatively few individuals [2, 9, 11]. Such sample sizes reduce statistical power and make estimates of dimensional accuracy, fit, deformation, or mechanical response more sensitive to specimen-to-specimen and participant-to-participant variability. In comparative studies of SLA and other AM technologies, limited sample sizes may also increase the risk of Type II errors and reduce confidence in claims of technological superiority. Consequently, findings from small experimental samples should be interpreted as preliminary evidence rather than definitive evidence for industrial adoption. Future studies should incorporate adequately powered sample sizes, repeated measurements, and appropriate statistical methods to improve reproducibility and generalizability across footwear geometries, materials, and user populations.

4.3 Lack of Longitudinal Studies

The limited availability of longitudinal studies restricts understanding of the stability and durability of SLA prototypes over extended periods. Existing investigations often rely on short-term mechanical or cyclic testing, which may not adequately characterize time-dependent phenomena such as creep, fatigue, moisture-related effects, or UV-induced degradation in photopolymer materials [6]. Consequently, short-term prototype performance should not be interpreted as evidence of equivalence to production footwear materials. Longer-term studies combining experimental testing with predictive modelling could provide better insight into material degradation, dimensional stability, and the implications of repeated loading for design and tooling decisions [4, 17]. Establishing standardized durability protocols and application-relevant exposure conditions would therefore strengthen the evidence base and improve confidence in the translational use of SLA for footwear pre-production.

4.4 Absence of Population-Specific Data

An important limitation of the current literature is the limited representation of population-specific foot anthropometry in SLA-based footwear research. Existing studies have frequently relied on relatively homogeneous samples, which may limit the generalizability of findings across populations with different foot dimensions, proportions, and morphological characteristics [5, 9, 13]. This is particularly relevant to footwear development in India, where variation in foot morphology may occur across sex, age, geographic regions, and individual anthropometric characteristics. However, population-level differences should be established through appropriately designed anthropometric studies rather than assumed from broad demographic classifications. The limited inclusion of diverse foot morphologies may affect the interpretation of fit, dimensional conformity, and pressure-distribution outcomes. Individuals with distinct anatomical characteristics, such as high or low medial arches, different forefoot widths, or variations in heel morphology, may interact differently with a given footwear geometry [8, 9, 12]. Consequently, findings derived from a narrowly defined sample may not adequately represent the performance of the same design across a broader user population. This limitation is particularly important for personalized footwear, where accurate representation of individual or population-specific anthropometric characteristics is central to design validation. Incorporating diverse foot-scan and anthropometric datasets could improve the assessment of last geometry, upper-foot conformity, sole dimensions, and other design parameters during SLA-based prototyping [3]. Future research should therefore prioritize sex- and age-diverse samples and systematically characterize relevant anthropometric variation, while including clinically or functionally distinct foot morphologies where appropriate. Such population-specific evidence would strengthen the translational relevance of SLA as a “design-validation and personalization tool” and support more inclusive development of footwear intended for diverse Indian users.

5 Implications

5.1 Design Implications

From an engineering perspective, resin-based SLA can support footwear design by providing high-resolution physical prototypes for early assessment of geometry, manufacturability, surface characteristics, and component interfaces. Prototypes can be used to identify potential issues associated with features such as undercuts, draft requirements, wall thicknesses, assembly clearances, and bonding interfaces before investment in production tooling [4, 14]. These assessments can complement digital manufacturability analysis and provide an opportunity for iterative design modification at relatively early stages of development. The high surface quality achievable with SLA can also facilitate evaluation of aesthetic features, textures, branding elements, and fine geometric details, which may be particularly relevant to footwear requiring precise visual and dimensional representation [2]. Similarly, SLA can be used to physically assess complex lattice or ribbed geometries intended to modify structural characteristics. However, geometric reproduction of such structures should not be interpreted as evidence of improved durability, cushioning, or energy return. These functional characteristics require subsequent material-specific mechanical testing under relevant loading conditions. Accordingly, SLA is best considered a “design-verification and de-risking tool within the pre-production workflow”, rather than a direct indicator of final product performance. Standardized reporting of resin formulation, printing parameters, build orientation, layer thickness, and post-processing conditions is necessary to improve reproducibility and confidence in design decisions [1, 3]. This approach may facilitate more efficient development of footwear concepts, including designs intended for improved fit, resource efficiency, or personalization, while subsequent functional and user-based evaluations remain necessary before claims regarding product performance can be established.

5.2 Occupational Health Relevance

Footwear used in occupational environments may require specific design characteristics related to fit, dimensional stability, traction, impact management, and user comfort. Resin-based SLA may contribute to the early development and assessment of such footwear by enabling high-resolution physical prototypes through which design features and component interfaces can be examined before production tooling [9, 14]. For example, prototype geometries may be used to assess the form and placement of outsole tread patterns, sole profiles, or interface features intended for subsequent testing of traction or impact-related performance. SLA prototypes may also support preliminary fit and ergonomic assessments by allowing footwear geometries to be evaluated against representative foot dimensions or three-dimensional foot-scan data [5, 11]. Such assessments can help identify geometric discrepancies and guide design modifications before fabrication of production components. However, a prototype-based assessment of geometry or fit does not establish that the resulting footwear will prevent slips, reduce fatigue, or decrease the risk of musculoskeletal disorders in occupational settings. Importantly, evidence linking footwear characteristics to occupational health outcomes requires appropriately designed biomechanical, ergonomic, longitudinal, and, where applicable, clinical or field-based studies. The current evidence base provides limited support for directly attributing health outcomes to SLA-based prototyping. Population diversity is also important because occupational footwear

may be used by individuals with substantial variation in foot morphology, body characteristics, work tasks, and exposure conditions [8]. Therefore, the principal occupational-health relevance of SLA lies in its potential to “support early design verification and iterative development of occupational footwear”, rather than to serve as evidence of a health intervention. Any claims regarding slip prevention, reduction of foot fatigue, musculoskeletal injury prevention, or improved occupational health should be based on subsequent validation of the final footwear product under representative occupational conditions [6, 12].

An important evidence gap concerns the limited availability of longitudinal studies examining the relationship between footwear design, plantar pressure distribution, and longer-term functional or injury-related outcomes. Short-term pressure measurements can provide useful information about how a footwear design influences load distribution, but they do not by themselves establish whether a particular design reduces injury risk or improves long-term musculoskeletal health. Consequently, SLA-based prototypes should be regarded as tools for early geometric and design validation, while evidence concerning injury prevention requires subsequent evaluation of the final footwear product using appropriately designed biomechanical, longitudinal, and clinical studies.

For translation into industrial practice, future studies should report printer characteristics, resin formulation, build orientation, layer thickness, post-processing conditions, dimensional accuracy, and relevant mechanical properties using standardized protocols. Such reporting would improve reproducibility and facilitate meaningful comparison between SLA systems and alternative AM technologies. In the Indian footwear context, integration with representative anthropometric and three-dimensional foot-scan datasets could further improve the applicability of this workflow to diverse user populations. Thus, SLA should be regarded as a “high-fidelity pre-production validation tool that reduces uncertainty between digital footwear concepts and production-ready designs”, while final claims regarding durability, comfort, biomechanical performance, injury prevention, or clinical effectiveness are dependent on subsequent validation of the finished footwear.

6 Future Research Directions

Future research should prioritize methodological standardization to improve the reproducibility and comparability of resin-based SLA studies in footwear development. Studies should systematically report and, where possible, standardize key parameters, including resin formulation, layer thickness, build orientation, printing conditions, post-curing duration and intensity, and post-processing procedures. Such standardization would facilitate more reliable comparison of dimensional accuracy and mechanical properties across different SLA systems and footwear geometries [4, 6]. Larger and more diverse experimental samples are also required to strengthen the statistical robustness and generalizability of existing evidence. Future studies should include multiple footwear geometries, resin systems, and representative participant populations, with appropriate sample-size justification and standardized testing protocols. Particular attention should be given to variation in foot anthropometry across sex, age, and other relevant morphological characteristics when evaluating fit and design conformity [5, 9, 13]. Longer-term mechanical and fatigue studies should further investigate the durability and dimensional stability of SLA prototypes under application-relevant loading and environmental conditions. Repeated-loading experiments, supported where appropriate by accelerated ageing and predictive modelling, could provide greater insight into fatigue behaviour, creep, and changes in mechanical properties over time. Importantly, such testing should distinguish photopolymer prototype behaviour from that of production footwear materials. Artificial intelligence and machine-learning approaches represent a further opportunity for process optimization. Models trained on systematically collected relationships between printing parameters, resin characteristics, geometry, and measured prototype outcomes could assist in predicting suitable layer thickness, build orientation, and post-curing conditions for specific footwear geometries. However, development of such models will require sufficiently large, standardized, and high-quality datasets before reliable implementation can be established. Finally, integration of large and representative three-dimensional foot-anthropometric databases with digital last development and SLA prototyping could support more individualized and inclusive footwear design [13]. Collaborative datasets involving diverse populations would improve external validity and provide a stronger foundation for evaluating personalized designs. Together, these priorities would help move SLA research from isolated proof-of-concept studies toward “standardized, reproducible, application-relevant pre-production validation frameworks”.

7 Conclusion

The present narrative review examined the role of resin-based SLA in pre-production physical prototyping for footwear development, with emphasis on design verification, manufacturability assessment, and early-stage risk reduction. The available evidence indicates that SLA can produce relatively high-resolution physical representations of complex footwear geometries, supporting the assessment of form, dimensional characteristics, surface features, fit-related geometry, and component interfaces before progression to production tooling. Its capacity to reproduce intricate geometries and fine surface details may be particularly valuable for iterative evaluation of complex sole structures and aesthetic features. SLA-based prototypes can also facilitate preliminary

manufacturability assessment, including evaluation of dimensional conformity, clearances, undercuts, assembly interfaces, and bonding concepts. However, these applications should be interpreted primarily as engineering and design validation and should not be equated with validation of the mechanical, biomechanical, clinical, or injury-prevention performance of final footwear. Differences between photopolymer resins and production materials may substantially influence flexibility, fatigue behaviour, creep, and long-term durability. The current evidence remains constrained by methodological heterogeneity, small sample sizes, limited population diversity, inconsistent reporting of printing and post-curing conditions, and a lack of long-term mechanical and fatigue testing. Future research should therefore prioritize standardized experimental protocols, adequately powered and diverse studies, application-relevant longitudinal testing, and validated computational approaches, including machine-learning-assisted process optimization. Integration of representative three-dimensional foot-anthropometric datasets may further support inclusive and personalized footwear development.

Author Contributions

Conceptualization, A.M.; methodology, A.M.; validation, A.M.; formal analysis, A.M.; investigation, A.M., T.J., C.R., and S.A.; data curation, T.J., C.R., and S.A.; review of literature, T.J., C.R., and S.A.; writing—original draft preparation, A.M.; writing—review and editing, A.M.; visualization, A.M.; supervision, A.M. All authors have read and agreed to the published version of the manuscript.

Data Availability

Not applicable.

Conflicts of Interest

The authors declare no conflicts of interest.

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